

Jamal Ud Din

Physics Teacher | Physics & Materials Science Researcher

Xi'an, Shaanxi, China | Open to Relocation (Saudi Arabia)

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PROFESSIONAL SUMMARY

Physics graduate and experienced educator seeking to bring a strong foundation in classical and applied physics into the secondary school classroom in Saudi Arabia. Combines a B.S. in Physics with active graduate research in materials science to deliver engaging, practical, and mathematically rigorous instruction. Proven track record in effective classroom management, facilitating laboratory demonstrations, and thoroughly preparing students for board examinations.

EDUCATION

Northwestern Polytechnical University

Xi'an, China

M.S. in Materials Science

2024 – Present

- Research focus: Combining experimental research and finite element analysis (FEA) to optimize diffusion bonding of Ti-60 titanium alloys for aerospace applications

University of Malakand

Khyber Pakhtunkhwa, Pakistan

B.S. in Physics

August 2018 – August 2022

- Core coursework: Classical Mechanics, Electromagnetism, Thermodynamics, Quantum Physics, Optics, Mathematical Physics

TEACHING EXPERIENCE

Physics Teacher

Government High and Secondary School Doagdara

August 2023 – July 2024

- Delivered comprehensive physics curricula and facilitated practical laboratory demonstrations to translate theoretical principles into hands-on learning
- Assessed student progress through regular examinations, providing targeted feedback to consistently improve academic performance
- Cultivated a disciplined, supportive classroom environment to successfully prepare students for final board examination requirements

RESEARCH EXPERIENCE

Graduate Researcher – Diffusion Bonding of Titanium Alloys

Northwestern Polytechnical University

2024 – Present

- Investigate diffusion bonding mechanisms of Ti-60 titanium alloys by combining experimental testing with Abaqus FEA simulations
- Analyze critical process parameters (temperature, pressure, time) to optimize joint reliability and microstructural integrity for high-performance applications
- Utilize Python scripting to automate simulation workflows and enhance data extraction efficiency for complex thermal-mechanical models

SKILLS

Physics & Science Knowledge: Classical Mechanics, Thermodynamics, Electromagnetism, Quantum Physics Fundamentals, Optics, Materials Science, Solid-State Physics, Applied & Engineering Physics

Research & Technical Skills: Finite Element Analysis (FEA), Abaqus Simulation, Python Engineering Automation, Diffusion Bonding Processes, Experimental Design, Materials Characterization, Data Analysis, Scientific Writing

Computer Skills: Python (scripting and automation), Microsoft Office (Word, Excel, PowerPoint), Data Visualization

Teaching & Interpersonal Skills: Lesson Planning & Curriculum Delivery, Classroom Management, Mentoring, Cross-Cultural Communication, Exam Preparation, Problem-Solving

LANGUAGES

English – Fluent (Professional Working Proficiency) | Arabic – Conversational